HONSHU CHEMICAL INDUSTRY CO.,LTD.
HONSHU CHEMICAL INDUSTRY CO., LTD.
Yaesu-daibiru Bldg, 1-1, Kyobashi, 1-Chome Chuo-ku, Tokyo 104-0031 Japan
Tel. 81-3-3272-1481 Fax. 81-3-3272-1480

SAFETY DATA SHEET

SECTION 1. IDENTIFICATION

COMPANY NAME : Honshu Chemical Industry Co., Ltd.
ADDRESS : Yaesu-Daibiru Building
1-1,Kyobashi 1-Chome, Chuo-ku, Tokyo 104-0031, Japan
DEPARTMENT : Specialty Chemicals Dept.
TELEPHONE No. : +81-3-3272-1481
TELEFAX No. : +81-3-3272-1480
EMERGENCY TELEPHON No. : Wakayama Works
+81-73-422-8171 24 hours Everyday
ADDRESS : 5-115 Kozaika 2-chome, Wakayama 641-0007, Japan

PRODUCT NAME : H-BHT
SDS No. : 012400
Ninth revised edition : 11 April 2018
GENERAL USE & PRODUCT DESCRIPTION : Antioxidants
Industry use only

SECTION 2. HAZARDS IDENTIFICATION

GHS : CLASSIFICATION

PHYSICAL HAZARDS :	**Hazard category**
EXPLOSIVES	Not applicable
FLAMMABLE GASES	Not applicable
AEROSOLS	Not applicable
OXIDIZING GASES	Not applicable
GASES UNDER PRESSURE	Not applicable
FLAMMABLE LIQUID	Not applicable
FLAMMABLE SOLIDS	Classification not possible
SELF-REACTIVE CHEMICALS	Not applicable
PYROPHORIC LIQUIDS	Not applicable
PYROPHORIC SOLIDS	Not classified
SELF-HEATING CHEMICALS	Classification not possible
WATER-REACTIVE FLAMMABLE CHEMICALS	Not applicable
OXIDIZING LIQUIDS	Not applicable
OXIDIZING INDIVIDUAL	Not applicable
ORGANIC PERROXIDE	Not applicable
METAL CORROSIVE SUBSTANCE	Classification not possible
DESENSITIZED EXPLOSIVES	Not applicable
HEALTH HAZARDS :	
ACUTE TOXICITY (Oral)	Classification not possible
ACUTE TOXICITY (Dermal)	Classification not possible
ACUTE TOXICITY (Inhalation Gases)	Not applicable
ACUTE TOXICITY (Inhalation Vapors)	Classification not possible
ACUTE TOXICITY (Inhalation Dusts and Mists)	Classification not possible
SKIN CORROSION / IRRITATION	Not classified
SERIOUS EYE DAMAGE / EYE IRRITATION	Category 2B
SENSITIZATION — RESPIRATORY	Classification not possible
SENSITIZATION — SKIN	Classification not possible
GERM CELL MUTAGENICITY	Classification not possible
CARCINOGENICITY	Classification not possible
REPRODUCTIVE TOXICITY	Category 2
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE)	Category 1 (nervous system)
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE)	Category 2 (lung, liver)
ASPIRATION HAZARD	Classification not possible
ENVIRONMENTAL HAZARDS :	
HAZARDOUS TO THE AQUATIC ENVIRONMENT —ACUTE TOXICITY	Category 1
HAZARDOUS TO THE AQUATIC ENVIRONMENT —CHRONIC TOXICITY	Category 1
HAZARDOUS TO THE OZONE LAYER	Classification not possible

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PICTOGRAM / SYMBOL :

SIGNAL WORD : Danger

HAZARD STATEMENT :
H320 Causes eye irritation
H361 Suspected of damaging fertility or the unborn child
H370 Causes damage to organs (nervous system)
H373 May cause damage to organs through prolonged or repeated exposure (lung, liver)
H400 Very toxic to aquatic life
H410 Very toxic to aquatic life with long lasting effects

PRECAUTIONARY STATEMENT :

【Prevention】
P264 Wash face and hands thoroughly after handling.
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understand.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P260 Do not breathe dust/fume/gas/mist/vapours/spray.
P270 Do not eat, drink or smoke when using this product.
P273 Avoid release to the environment.

【Response】
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337+P313 IF eye irritation persists: Get medical advice/attention.
P308+P313 IF exposed or concerned: Get medical advice/attention.
P308+P311 IF exposed or concerned: Call a POISON CENTER/doctor.
P321 Specific treatment (see § 4,11 on this label).
P314 Get medical advice/attention if you feel unwell.
P391 Collect spillage.

【Storage】
P405 Store locked up.

【Disposal】
P501 Dispose of contents / container in accordance with local / regional / national / international regulation.

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SECTION 3. COMPOSITION / INFORMATION ON INGREDIENTS

Substance/mixture : Substance

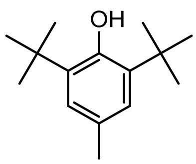
《Main component》

CHEMICAL NAME : 2,6-Bis(1,1-dimethylethyl)-4-methylphenol

SYNONYMS :
2,6-Di-tert-butyl-p-cresol
2,6-Di-tert-butyl-4-cresol
2,6-Di-tert-butyl-4-methylphenol
Butylated hydroxytoluene
Reaction product from p-cresol and isobutylene
Phenol, 2,6-bis(1,1-dimethylethyl)-4-methyl-
BHT

PURITY : More than 99.0 %
FORMULA : C15 H24 O
CAS No. : 128-37-0
TSCA : Listed on inventory.
EC No. : Listed on inventory. No. 204-881-4

Structural Formula



《Impurity》
By-product and others : Less than 1.0%

SECTION 4. FIRST-AID MEASURES

SKIN CONTACT : Immediately wash materials off the skin with soap and copious amounts of water.
If redness, itching or a burning sensation developed, get medical attention.
Wash contaminated clothing and decontaminate footwear before reuse.

EYES CONTACT : Immediately flush with copious amounts of water for at least 15 minutes. If redness, itching or burning developed, have eyes examined and treated by medical personnel.

INGESTION : If swallowed, dilute water.
DO NOT INDUCE VOMITING.
Never give fluids or induce vomiting if the victim is unconscious or having convulsions.
(never give anything by mouth to an unconscious person.)
When vomiting occurs, keep head lower than hips to help prevent aspiration. If person is unconscious, tern head to side. Get immediate medical attentions.

INHALATION : Remove victim to fresh air. If cough or other respiratory symptoms developed, consult medical personnel. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. And then, consult medical personnel.

POTENTIAL HEALTH EFFECTS : See Section 11 for more information.

SECTION 5. FIRE-FIGHTING MEASURES

FIRE AND EXPLOSION HAZARDS : Slight fire hazard. Dust/air mixtures may ignite or explode.

EXTINGUISHING MEDIA : Water fog, foam, carbon dioxide, drychemicals. All extinguishing materials are suitable. Fire-men have to wear self-contained breathing apparatus. Be sure to wear protective gear during fire-fighting.

LARGE FIRES : Use regular foam or flood with fine water spray.
It is effective to use foam to cut off oxygen supply to the fire.

FIRE FIGHTING : Move container from fire area if it can be done without risk. Do not scatter spilled material with high-pressure water streams. Dike for later disposal. Use extinguishing agents appropriate for surrounding fire. Avoid inhalation of material or combustion by-products. Stay upwind and keep out of low areas.
Containers can rupture and release highly toxic vapors or decomposition products if exposed to heat.
Closed containers could violently rupture. Cool container with water spray.
Be sure to wear protective gear during fire-fighting

PROTECTIVE EQUIPMENT : During emergency conditions, overexposure to thermal decomposition products may cause health hazard. Self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes should be worn.

UNUSUAL FIRE AND EXPLOSION : Formation of phenol, phenol-derivatives, in case of fire or during thermal decomposition. Dust, like most organic materials in powder form , may form explosive mixture with air.

DECOMPOSITION PRODUCTS : Thermal decomposition may produce phenols, carbon monoxide,carbon dioxide.

PRODUCT OF COMBUSTION : Oxides of carbon

SECTION 6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS : Use personal protection recommended in Section 8.
Isolate the hazard area and deny entry to unnecessary and unprotected personnel.

ENVIRONMENTAL PRECAUTIONS : Keep out of drains, sewers, ditches and waterways.
Minimize use of water to prevent environment contamination.
Do not let spilled or leaking material enter waterways.

METHODS OF CLEAN-UP : Use spark-proof tools (explosion-proof equipment) to sweep or scrape up and containerize.
Dust can be a fire or explosion hazard.

Large Leak : Use a safety spark-free shovel to scoop the leak and put it in a container.

Small Leak : Wipe it off with rag and put it in a container.

OCCUPATIONAL RELEASE : Collect spilled material in appropriate container for disposal. Keep out of water supplies and sewers. Keep unnecessary people away, isolate hazard area and deny entry.

PRECAUTIONS IN CASE OF SPILLAGE OR LEAKAGE : Ventilate area and wash spill site after material pick up is complete.
Wear respirator, chemical safety goggles, impervious boots and chemical-resistant gloves.. Sweep up, place in a bag and hold for waste disposal. Avoid raising dust and eliminate all ignition sources.

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SECTION 7. HANDLING AND STORAGE

Store and handle in accordance with all current regulations and standards.

HANDLING : Do not breathe dust.
Minimize dust generation and accumulation.
Chemical safety goggles. Chemical-resistant gloves. OSHA/MSHA-approved respirator.
Safety shower and eye bath. Mechanical exhaust required. Avoid contact and inha
and inhalation. Do not get in eyes, on skin, on clothing. Keep container tightly closed
irritating dust. Use with adequate ventilation. Avoid raising dust eliminate all
ignition sources. In filling operations take precautionary measures against static
discharges. To avoid sudden release of pressure, loosen closure cautiously before
opening.

STORAGE : Store in a cool, dark, dry place and in a well-ventilated area.
Store in closed containers away from moisture, heating and dusting.
Keep separated from incompatible substances.
Keep container closed when not in use.
After opening, use as soon as possible.

INCOMPATIBILITY : See Section 10 for more information.

CONDITIONS TO AVOID : Avoid heat, flames, sparks and other sources of ignition.
Avoid generating dust. Avoid contact with incompatible materials.

SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

EXPOSURE LIMIT AT THE WORK PLACE :

PRODUCT : No occupational exposure limits established.

NUISANCE PARTICULATES (NUISANCE DUST) ;

OSHA TWA	5mg/m3	(respirable dust fraction)
OSHA TWA	15mg/m3	(total dust)
ACGIH TWA	10mg/m3	(inhalable particulate)
ACGIH TWA	3mg/m3	(respirable particulate)
DFG MAK	4mg/m3	(inhalable dust fraction)
DFG MAK	1.5mg/m3	(respirable dust fraction)

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VENTILATION : Provide local exhaust ventilation system. Ventilation equipment should be explosionresistant if explosive concentrations of material are present. Ensure compliance with applicable exposure limits.

PROTECTIVE MEASURE : An appropriate respirator for dust should be worn. Chemical resistant gloves should be put on. Safety glasses should be worn in any type of industrial operation.

TECHNICAL PROTECTIVE MEASURES : Safety shower and eye bath. Mechanical exhaust required.

Keep away from foodstuffs, drinks and tobacco. Wash hands before breaks and at end of work.
Keep working clothes separate. Take off immediately all contaminated clothing.
Avoid inhaling dust. After contact with skin, wash immediately with plenty of water and soap.

RESPIRATOR : Under conditions of frequent use or heavy exposure, respiratory protection may be needed. Respiratory protection is ranked in order from minimum to maximum. Consider warning properties before use.
Any dust, mist, and fume respirator.
Any air-purifying respirator with a high-efficiency particulate filter.
Any powered, air-purifying respirator with a dust, mist, and fume filter.
Any powered, air-purifying respirator with a high-efficiency particulate filter.

For Unknown Concentrations or Immediately Dangerous to Life or Health:
Any supplied-air respirator with full facepiece and operated in a pressuredemand or other positive-pressure mode in combination with a separate escape supply.
Any self-contained breathing apparatus with a full facepiece

There are not known hazards associated with this material when used as recommended.
The following general hygiene considerations are recognized as common good industrial hygiene practices.

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SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE (color, physical form, shape etc.)	:	White to pale yellowish crystalline powder
PRODUCT ODOR	:	Odorless to slight alkylphenol-like (sweet odor)
PHYSICAL STATE	:	Solid (at 20℃)
pH	:	No Data
MOLECULAR (FORMULA) WEIGHT	:	220.4
MELTING / FREEZING POINT (specify which)	:	>69.2℃
INITIAL BOILING POINT AND BOILING RANGE	:	265℃/0.1MPa(760mmHg)
FLASH POINT	:	127℃ (O.C.)
UPPER / LOWER FLAMMABILITY OR EXPLOSIVE LIMITS	:	LOWER 35g/m^3 (125～600 μm) 3000g/m^3
VAPOR PRESSURE	:	1.3Pa(20℃)
SPECIFIC GRAVITY (water＝1) OR DENSITY	:	890g/L (at 80℃) 1.048 (20/4℃)
BULK DENSITY	:	0.620～0.640 g/ml
SOLUBILITY	:	Water : Insoluble 0.4mg/L(at 20℃) 0.6mg/L(at 25℃) Methanol : Soluble Acetone : Soluble Toluene : Soluble
DISSOCIATION CONSTANT pKa	:	12.2
OCTANOL / WATER PARTITION COEFFICIENT logPow	:	5.1
AUTO-IGNITION TEMPERATURE	:	>640℃
DECOMPOSITION TEMPERATURE	:	No Data
HEAT OF COMBUSTION	:	42.48kJ/g(10.148kcal/g)
MINIMUM IGNITION ENERGY	:	1mJ

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SECTION 10. STABILITY AND REACTIVITY

REACTIVITY	:	No data
CHEMICAL STABILITY	:	Stable under normal condition.
POSSIBILITY OF HAZARDOUS REACTIONS	:	Under dusty condition, may form flammable/explosive dust-air mixtures.
CONDITIONS TO AVOID	:	Avoid heat, flames, sparks and other sources of ignition. Minimize contact with dusts of material. Avoid inhalation of dust.
INCOMPATIBILE MATERIALS	:	Bases, reducing agents, acid chlorides, acid anhydrides and oxidizing agents. Corrodes steel, brass, copper, copper alloys and aluminum.
HAZARDOUS DECOMPOSITION PRODUCTS	:	Will not polymerize under normal condition.
HYDROLYSIS	:	Will not hydrolyze under normal condition.

SECTION 11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY	:	Woman	(oral)	TDLo		80mg/kg	
		RATS	(oral)	LD50		1700-1900mg/kg	
			(oral)	LD50		2450mg/kg	
			(oral)	LD50		>2930mg/kg	
			(oral)	LD50		>10000mg/kg	
			(skin)	LD50		>2000mg/kg	
SKIN CORROSION / IRRITATION	:	HUMAN	(skin)		500mg/48hrs		MILD
		RABBIT	(skin)		500mg/48hrs		MODERATE
SERIOUS EYE DAMAGE / EYE IRRITATION	:	RABBIT	(eye)		100mg/24hrs		MODERATE
		NITE classfied category2B					
SENSITIZATION — RESPIRATORY	:	No data					
SENSITIZATION — SKIN	:	No data					
GERM CELL MUTAGENICITY	:	Negative in the Ames test.					
CARCINOGENICITY	:	IARC Cancer Review Group 3					
		ACGIH A4 (Not Classifiable as a human Carcinogen)					
REPRODUCTIVE TOXICITY	:	NITE classfied category2					
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE)	:	NITE classfied category1					
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE)	:	NITE classfied category2					
ASPIRATION HAZARD	:	No data					

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SECTION 12. ECOLOGICAL INFORMATION

Avoid release to the environment.
Do not empty into drains or water.

BIODEGRADABILITY : Non-biodegradability
Test Equipment (Standard type) Test period (4weeks)
Chemical Concentration (50ppm),
Concentration of Activated Sludge (50ppm)
Indirect Analysis : 4.5% (by BOD)
Direct Analysis : 0.8% (by HPLC)

BIOCONCENTRATION : Moderate-bioconcentration
Test Equipment (Standard type) Test period (8weeks)
Species (Carp [Cyprinus carpio])

1st Concentration area	: (500ppb)	BCF	:	220-2800
2nd Concentration area	: (50ppb)	BCF	:	230-2500
3rd Concentration area	: (5ppb)	BCF	:	330-1800

The test was completed in 6 weeks for 1st concentration area. An abnormality of the back was found in 4 tails of 14 tails only in 1st concentration area.

ENVIRONMENTAL TOXICITY:

FISH TOXICITY :				
Acute Toxicity	Rice fish (Oryzias latipes)	TLm(LC50)	48hrs	5.0ppm(w/v)
	Oryzias latipes	LC50	48hrs	5.3mg/L
		LC50	96hrs	1.1mg/L
INVERTEBRATE TOXICITY :				
Acute immobilization	Daphnia magna	EC50	48hrs	0.84mg/L
Reproduction	Daphnia magna	EC50	21days	0.096mg/L
		NOEC	21days	0.069mg/L
ALGAL TOXICITY :				
Growth inhibition	Selenastrum capricomutum			
	Growth rate	EC50	72hrs	>7.0mg/L
		NOEC	72hrs	1.7mg/L
	AUG	EC50	72hrs	>10mg/L
		NOEC	72hrs	1.0mg/L

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SECTION 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with federal, state, local and all applicable regulations.
Recover or recycle if possible.
Avoid release to the environment.
Do not empty into drains or water.

WASTE DISPOSAL : Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and a scrubber. Observe all federal, state and local regulations concerning health and environment when disposing of this substance. Dispose in accordance with all applicable regulations.

WASTE DISPOSAL METHOD : Waste must be disposed of in accordance with federal, state and local environmental control regulations. Incineration is the preferred method. Empty containers must be handled with care due to product residue.

DO NOT HEAT OR CUT EMPTY CONTAINER WITH ELECTRIC OR GAS TORCH.

CONTAINER DISPOSAL : Empty containers retain product residue. Observe all hazard precautions. Keep away from heat, sparks and flames. Do not distribute, make available, or reuse empty containers except for storage andshipment of original product. Remove all hazardous product residue, and puncture or otherwise destroy empty containers before disposal.

SECTION 14. TRANSPORT INFORMATION

Treat the container / package carefully.

UN No. : 3077 (ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S)
UN Hazardous class : Class 9 (Miscellaneous dangerous substances and articles)
UN Packing Group (Class) : Ⅲ
LAND TRANSPORT ADR / RID : Class 9 Packing Group Ⅲ
AIR TRANSPORT IATA / ICAO : Class 9 Packing Group Ⅲ
Ⅲ : PAT956(400kg), Y956(30kg), CAO956(400kg)
MARITIME TRANSPORT IMDG : Class 9 Packing Group Ⅲ
Environmental hazards : Not applicable.

SECTION 15. REGULATORY INFORMATION

Follow all regulations in your country.

TSCA	:	Listed on inventory.	
EC No.	:	Listed on inventory.	No. 204-881-4
KECL	:	Listed on inventory.	No. KE-03079
(N)DSL	:	Listed on inventory.	
AICS	:	Listed on inventory.	
IECSC	:	Listed on inventory.	
TCSI	:	Listed on inventory.	
NZIoC	:	Listed on inventory.	No. HSR002784
PICCS	:	Listed on inventory.	
SWISS	:	Listed on inventory.	No. G-2202
TECI	:	Not listed on inventory.	
ENCS	:	Listed on inventory.	No. 3-540 , 9-1805
ISHL	:	Listed on inventory.	

SECTION 16. OTHER INFORMATION

BE CAREFUL : Not completely substance tests.

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.
Please carry out safety and other tests in advance and judge the conformity to your intended use on your responsibility.

[NOTE]

The date of preparation of the revision.

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Third revised edition	:	3 September 2009
Fourth revised edition	:	6 October 2010
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Sixth revised edition	:	17 August 2012
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